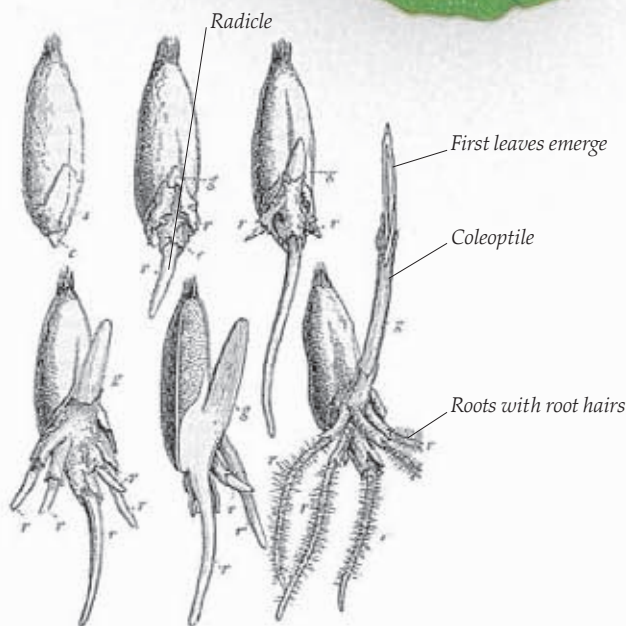


First pair of leaves,
now fully grown

Leaf stalk

GERMINATING GRAIN

Wheat is a monocot, which means that it has just one seed leaf (p. 9). The young shoot grows upward through the soil, protected by a tube called a coleoptile. As with all grasses, the growing point, or apical meristem, of the leaves of the wheat plant is at ground level and can continue to produce new shoots at the base of the plant, even if the leaves are removed. This is why pasture can survive being nibbled by cows, and why lawns thrive on being mown. Other plants, such as the bean, have their growing region, or meristem, near the tip. If they are mown or nibbled, they must grow out from side buds instead.



Radicle

First leaves emerge

Coleoptile

Roots with root hairs

Fully
upright
stem

Seed case is now
not needed and
starts to shrivel

4 THE RACE TO REPRODUCE

Once germination is complete, the scarlet runner grows quickly. Because it is a climbing plant, it does not need to develop thick stems. Instead, it uses other plants for support (p. 38). Given the right conditions, the plant will produce its first flowers in six weeks. After pollination (pp. 22–23) and fertilization, these will develop into pods full of seeds. By the time the seeds have dried out, the life cycle of the scarlet runner has turned full circle.

Leaves develop from
an underground
tuber

A YEARLY CYCLE

Some plants produce underground storage organs, such as bulbs, tubers, and corms (pp. 32–33). Every fall the leaves of lords-and-ladies die, but the following spring new leaves develop from buds on the tuber. Although this process may look very similar to germination, it is very different.



Thick mass of
roots absorbs
water and
nutrients
from the soil